

MILIND RANE

Cloud Infrastructure & Machine Learning Engineer

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SUMMARY

Cloud infrastructure engineer with 2+ years operating production AWS and Kubernetes platforms - Terraform, Amazon EKS, Helm, CI/CD, and observability - who also designs and trains models in PyTorch, with a published patent application in computer vision. Sustains 99.9% availability while cutting cloud costs 12%.

CLOUD INFRASTRUCTURE EXPERIENCE

Teleglobal International Pvt. Ltd.

August 2025 - Present

Cloud Infrastructure Engineer

- Operate a production Amazon EKS platform provisioned with Terraform and delivered through Helm with autoscaling, sustaining 99.9% availability.
- Automated Infrastructure as Code, CI/CD, and deployment workflows across Jenkins, GitHub Actions, Docker, and Amazon ECR, improving delivery efficiency 30%.
- Reduced AWS costs 12% through Well-Architected reviews - rightsizing compute, eliminating idle resources, and correcting infrastructure drift.
- Authored reusable Terraform modules with S3/DynamoDB remote state, making environment build-out a repeatable, version-controlled apply.
- Own incident response and root-cause analysis, diagnosing production failures from Prometheus, Grafana, and CloudWatch telemetry.

Kali IO Solutions LLP

January 2025 - July 2025

Technical Support & Operations Consultant

- Delivered production AWS deployments and application rollouts for enterprise customers, executing migrations with zero downtime.
- Resolved infrastructure and backend API failures in production, restoring service and closing recurring deployment defects.
- Advised customers on cloud deployment planning and implementation.

Grit Pulse

June 2024 - December 2024

Python Developer Intern

- Built the LunaPilot AI reinforcement learning agent and e-Pharmacy hybrid recommender in Python, covering model design, training, and evaluation.

MACHINE LEARNING

Lip Reading for Hindi and Marathi Languages using Deep Learning | Patent Application Published

Indian Patent Office, Application No. 202521037574 A | Python, CNN, LSTM, Streamlit

- Developed a hybrid CNN-LSTM deep learning model for accurate lip-reading in Hindi and Marathi.
- Created a custom dataset with annotated phoneme sequences, addressing language-specific data gaps.
- Built a Streamlit interface for real-time video input and transcription output.

LunaPilot AI - Deep Q-Learning for Autonomous Lunar Landing

PyTorch, OpenAI Gym, reinforcement learning

- Trained a Deep Q-Network in PyTorch on OpenAI Gym's LunarLander-v2 using experience replay, target networks, and epsilon-greedy exploration, reaching a 250+ average episode score.
- Implemented a neural Q-function approximator to estimate optimal policies for the continuous control task.

Hybrid Recommendation System for e-Pharmacy

Python, NLP, scikit-learn, Surprise

- Combined content-based NLP (CountVectorizer, cosine similarity) with SVD collaborative filtering to build a hybrid medicine recommender.
- Improved recommendation relevancy 15% over a content-only baseline by merging semantic and behavioural signals.

INFRASTRUCTURE PROJECTS

AWS Cost & Security Automation Platform

Python, AWS Lambda, EventBridge, CloudWatch, Terraform

- Automated AWS governance with a Python serverless platform - Lambda, EventBridge, CloudWatch - provisioned end-to-end in Terraform.
- Detected idle resources, over-permissive IAM, public S3 exposure, and infrastructure drift, recovering unused spend and eliminating misconfigurations.

AI/ML Ready Kubernetes Platform

Amazon EKS, Docker, Terraform, Helm, Python, Prometheus, Grafana

- Provisioned a scalable Amazon EKS platform with Terraform for hosting containerized AI/ML inference services.
- Configured autoscaling, Prometheus and Grafana monitoring, centralized logging, and secured deployment pipelines for ML workloads.
- Packaged the stack as reusable infrastructure for repeatable AI platform deployments.

MLOps Infrastructure using AWS & Kubernetes

AWS, Terraform, Docker, Kubernetes, GitHub Actions, FastAPI

- Developed Terraform Infrastructure as Code supporting scalable model deployment workflows.
- Built GitHub Actions CI/CD pipelines automating containerized deployment of machine learning services.
- Integrated monitoring and logging across the deployment infrastructure.

EDUCATION & CERTIFICATIONS

Bachelor of Engineering, Information Technology

2025

Smt. Kashibai Navale College of Engineering, Savitribai Phule Pune University

Microsoft Certified: Azure Developer Associate (AZ-204) | Python Development Training - IIT Kharagpur

Training (AWS Skill Builder): Well-Architected Best Practices & Foundations, Partner Migration Foundations

TECHNICAL SKILLS

Machine Learning: PyTorch, deep reinforcement learning (DQN), NLP, scikit-learn, Surprise, collaborative filtering

Languages & Systems: Python, Bash, SQL, Linux (Ubuntu, Amazon Linux), PostgreSQL, MySQL

AWS: EC2, EKS, ECS, ECR, S3, IAM, VPC, Application Load Balancer, Route 53, Lambda, CloudWatch, EventBridge, Auto Scaling, DynamoDB

Infrastructure & Containers: Terraform (modules, remote state), CloudFormation, Docker, Kubernetes, Helm, HPA

CI/CD & Observability: Jenkins, GitHub Actions, Git/GitHub, SonarQube, Prometheus, Grafana, ELK Stack